

The empirical recurrence for OEIS A259249: recovering a conjecture that is not written down

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Abstract

OEIS A259249 records “Empirical recurrence of order 54”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 57$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 54, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 54 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A259249 is “Number of $(n+1) \times (7+1)$ 0..1 arrays with each 2×2 subblock having clockwise pattern 0000 0011 or 0111.”. It has offset 1 and begins

1353, 15613, 159168, 1803859, 18970267, 211569948, 2259371001, 24946035357, ...

The entry states:

Empirical recurrence of order 54 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:49:54, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 257$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 57$ of the 257, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 57$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 114$ exact terms of the model returns order 54 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{54} c_i a(n-i),$$

c_1	15	c_{15}	−9383006715272	c_{29}	142351179116683968	c_{43}	−1247575015462561
c_2	210	c_{16}	−13155509255208	c_{30}	680672797407900823	c_{44}	−702970163865606
c_3	−3678	c_{17}	104004223595670	c_{31}	−114538113520510003	c_{45}	159022749710136
c_4	−18786	c_{18}	168988056692059	c_{32}	−826294716435341094	c_{46}	29438821044104
c_5	394288	c_{19}	−837880339950595	c_{33}	51460449101764786	c_{47}	−9529199244720
c_6	998957	c_{20}	−1570115512498570	c_{34}	709513304519490804	c_{48}	−227934437184
c_7	−24792161	c_{21}	4852014206848964	c_{35}	−11776978902972452	c_{49}	233482213952
c_8	−38205170	c_{22}	10447315083641748	c_{36}	−425247111174642173	c_{50}	−11653522304
c_9	1027319042	c_{23}	−19896533353481060	c_{37}	6880435643035213	c_{51}	−1484766208
c_{10}	1211342424	c_{24}	−49498784510341195	c_{38}	173783031692632300	c_{52}	114800128
c_{11}	−29723021248	c_{25}	56789581095380513	c_{39}	−8448557251856204	c_{53}	373760
c_{12}	−33336762439	c_{26}	166636515670061546	c_{40}	−46596622453572640	c_{54}	−61440
c_{13}	618554705691	c_{27}	−110555675413124062	c_{41}	4723149546286386		
c_{14}	755387168384	c_{28}	−399023029981114782	c_{42}	7694240756734353		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 54, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $54 + 4$ terms removed returns order 54 as well, so $\deg q_{\min} = 54 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $54 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 54 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A259249.
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